AIR UNIVERSITY AEROSPACE AND AVIATION CAMPUS KAMRA

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COURSE: Digital Image Processing

Lab task 6

2. Objective

The objective of this lab is to develop a **Smart Image Enhancement System** that integrates multiple image processing techniques from previous labs, allowing users to:

1. Upload and visualize images with metadata.
2. Apply sampling, quantization, geometric transformations, intensity adjustments, and contrast enhancement.
3. Observe the effect of each transformation on image quality and histogram distribution.
4. Save the enhanced image and compare it with the original.

3. Methodology

Phase	Description
Phase 1: GUI Design	Using Tkinter to create a clean, modern interface with a scrollable sidebar, two image panels, and uniform grey buttons.
Phase 2: Image Upload & Metadata	Load images in OpenCV, display resolution, channel count, and datatype in the console.
Phase 3: Sampling & Quantization (Lab 02)	Implement resolution scaling and bit-depth reduction. Downsampling and upsampling applied for resolution changes.
Phase 4: Geometric Transformations (Lab 03)	Rotate the image using OpenCV's rotation matrix. Reversible transformations are preserved.
Phase 5: Intensity Adjustments (Lab 04)	Negative transformation and gamma correction applied to adjust brightness and contrast.
Phase 6: Contrast Enhancement (Lab 05)	Histogram equalization performed for improved contrast. RGB histogram visualizations added for analysis.
Phase 7: Saving & Comparison	Save enhanced images and compare with original for before/after analysis.

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4. Code Snippets

```
# Gamma Correction Example
gamma = self.gamma_val.get()
```

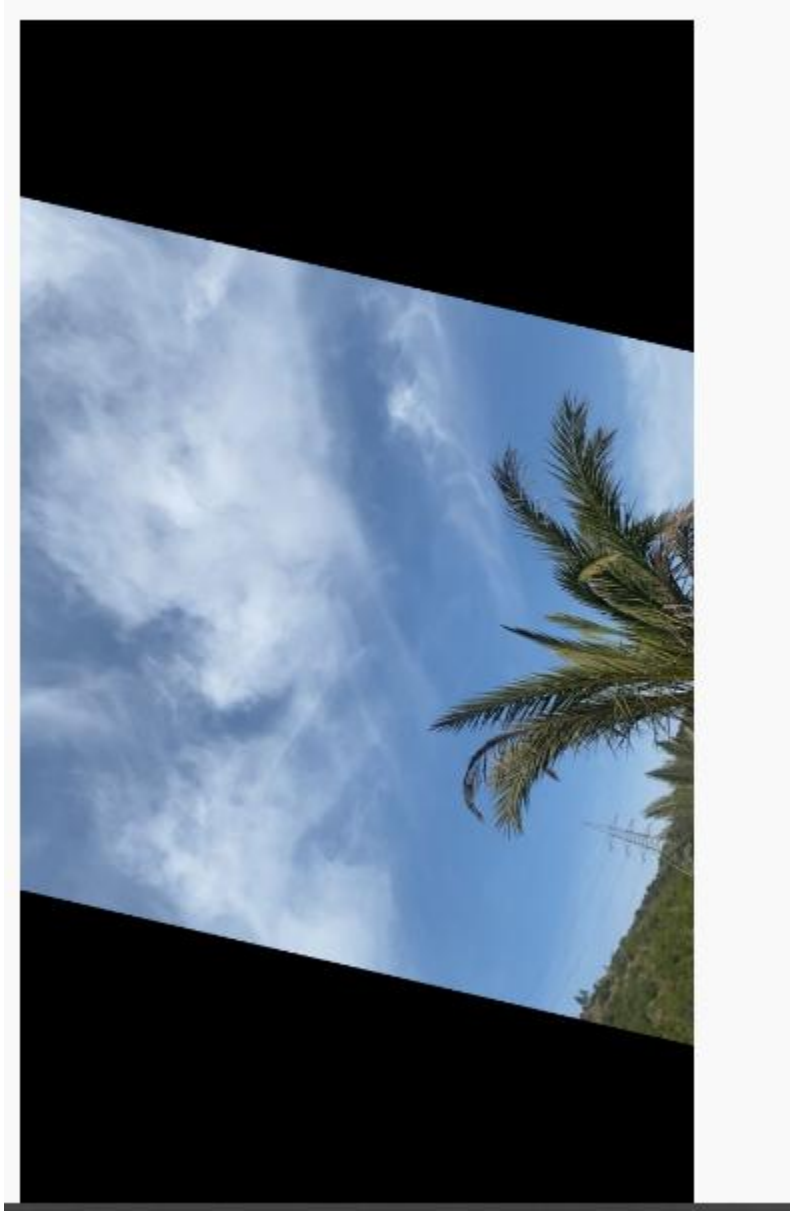
```
invGamma = 1.0 / gamma
table = np.array([((i / 255.0) ** invGamma) * 255 for i in range(256)], dtype='uint8')
self.processed_img = cv2.LUT(self.original_img, table)
self.show_on_panel(self.processed_img, self.panelB, is_primary=False)

# Histogram Equalization Example
ycrcb = cv2.cvtColor(self.original_img, cv2.COLOR_BGR2YCrCb)
ycrcb[:, :, 0] = cv2.equalizeHist(ycrcb[:, :, 0])
self.processed_img = cv2.cvtColor(ycrcb, cv2.COLOR_YCrCb2BGR)
```

5. Input/Output Images

- Include screenshots of:
    1. **Original Image**

2. **After Sampling & Quantization**

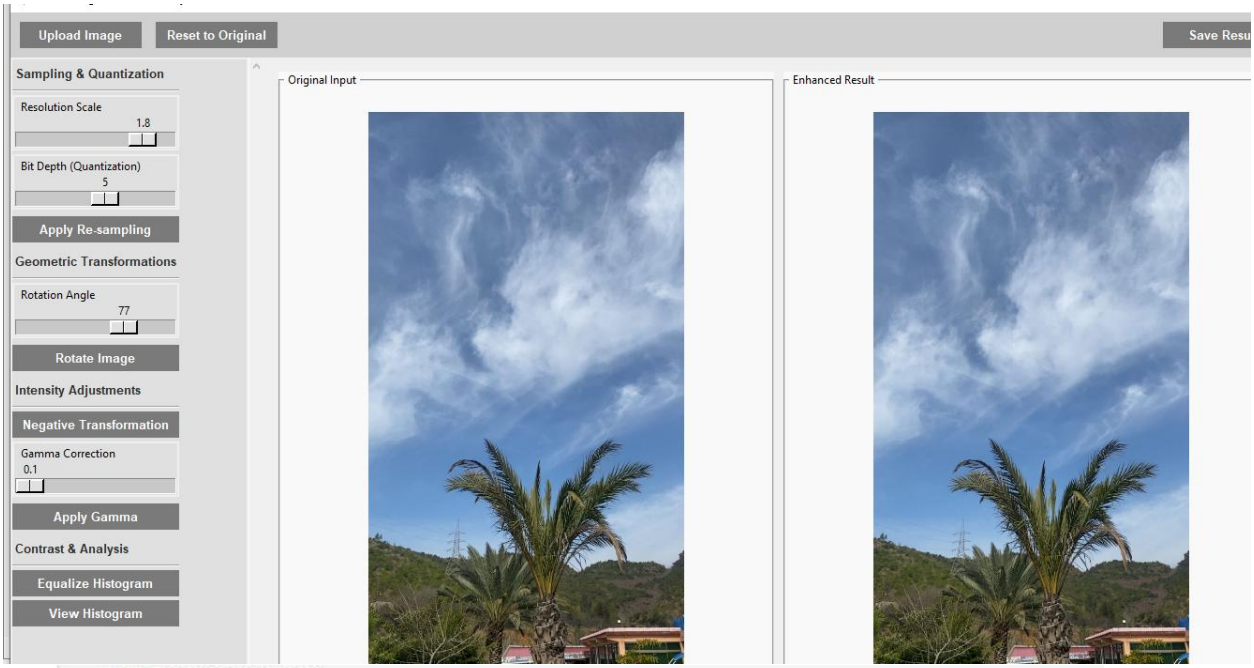
3. **After Rotation**



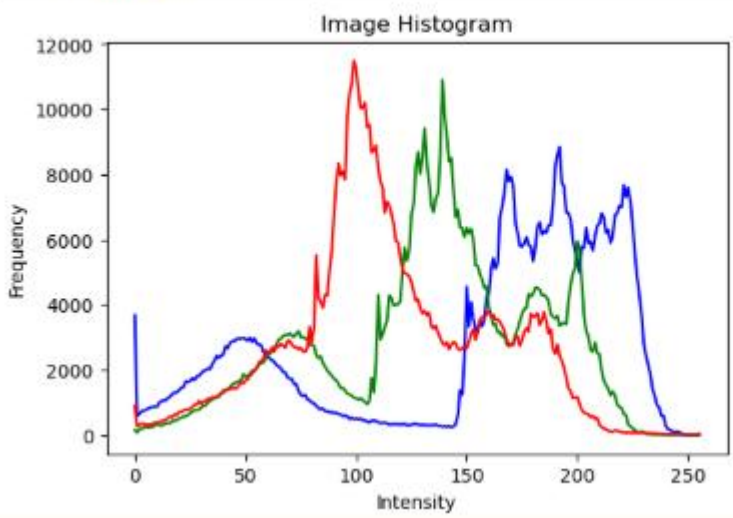
4. **After Negative Transformation**

5. **After Gamma Correction**

6. **After Histogram Equalization**




```
app = ImageSystem(root)
root.mainloop()
```



6. Observations & Analysis

1. **Sampling & Quantization:** Reducing bit depth reduces the number of gray levels, which can introduce visible banding.
2. **Rotation:** Maintains intensity values but changes spatial orientation.
3. **Negative Transformation:** Inverts intensity values, useful for enhancing dark details.

4. **Gamma Correction:** Brightness is adjusted non-linearly; gamma < 1 brightens, gamma > 1 darkens the image.
5. **Histogram Equalization:** Enhances contrast by redistributing intensity levels uniformly across the image.
6. **Overall:** Combining multiple enhancements produces a visually sharper and more informative image.

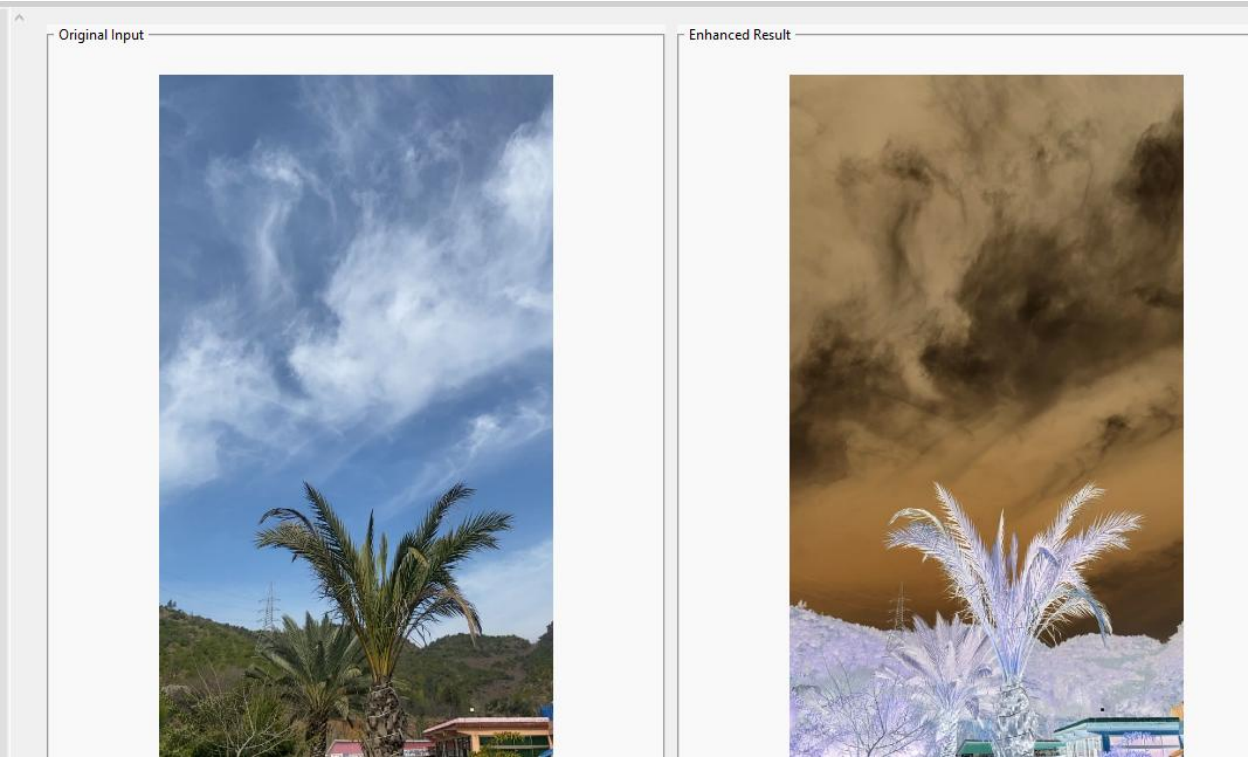
7. Conclusion

The **Smart Image Enhancement System** successfully integrates multiple lab tasks into a **single GUI-based tool**, allowing students to understand how **different image processing operations** affect image quality and histogram distribution.

- Histogram equalization and gamma correction were most effective in improving visibility and contrast.
- The project demonstrates practical application of theoretical image processing concepts.

8. LinkedIn Submission

Before/After Screenshot –



Key Learning Outcomes:

- Learned to integrate multiple image processing operations in Python GUI.
- Understood how transformations and quantization affect image quality.
- Gained experience in designing user-friendly interactive interfaces.

Instructor Mention: Mr. Ghulam Ali
LinkedInLink: https://www.linkedin.com/posts/rida-nayab-80189a339_python-opencv-tkinter-activity-7445809144298770432

Completed Lab 06: Smart Image Enhancement System!
Implemented a Python GUI using Tkinter and OpenCV with features like sampling, quantization, rotation, gamma correction, and histogram equalization.
Check out the GitHub repository: https://github.com/ridanayab145-rgb/Smart-Image-Enhancement-System
#Python #ImageProcessing #OpenCV #GUI #DigitalImageProcessing

https://www.linkedin.com/posts/rida-nayab-80189a339_python-opencv-tkinter-activity-7445809144298770432-6y3D?utm_source=share&utm_medium=member_desktop&rcm=ACoAAFUG0QEBjz5Z1mdpZilkzanZV9PzbL5Jzxc

9. Questions & Answers

1. **Why does histogram equalization improve contrast?**
   It redistributes the intensity levels to occupy the full range, making dark areas darker and bright areas brighter.

2. **How does gamma affect brightness?**
   Gamma correction adjusts intensity non-linearly. Gamma < 1 brightens the image, gamma > 1 darkens it.

3. **What is the effect of quantization on image quality?**
   Reducing bit depth reduces gray levels, which may cause loss of detail and visible banding.

4. **Which transformation is reversible and why?**
   Rotation (by multiples of 90°) and negative transformation are reversible because original pixel intensities can be recovered without information loss.

5. **How do transformations affect spatial structure?**
   Geometric transformations change pixel positions but usually maintain intensity. Some operations like rotation may introduce minor interpolation artifacts.